



UNIVERSITY
OF ŁOMŻA

DATABASE DESIGN

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Public Transport & Ticketing System

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1. Project Topic & Real-World Scenario

"CityLink" is an integrated management platform designed for urban public transport networks. The system is designed to synchronize vehicles (buses/trams) across various routes and manage the passenger ticketing process.

In this scenario, the transport authority manages multiple Routes, each consisting of several Stations (RouteStops). Drivers are assigned to Vehicles to perform scheduled Trips. Passengers can purchase Tickets, which are linked to their personal profiles and specific trips. The system also handles Payments, applies Discounts (e.g., for students), and logs Delays to improve service quality.

2. Functional & Non-Functional Requirements

Functional Requirements:

- **Passenger Management:** Registration of personal data (Name, Surname, Contact) and modification of profiles.
- **Route & Schedule Control:** Ability to create routes, assign stations in order, and schedule recurring trips.
- **Ticketing & Payments:** Generation of unique tickets for passengers and processing of payments with applicable discounts.
- **Fleet & Personnel Tracking:** Management of vehicle data and driver assignments to specific trips.
- **Monitoring:** Logging and displaying delays for specific trips to inform passengers.

Non-Functional Requirements:

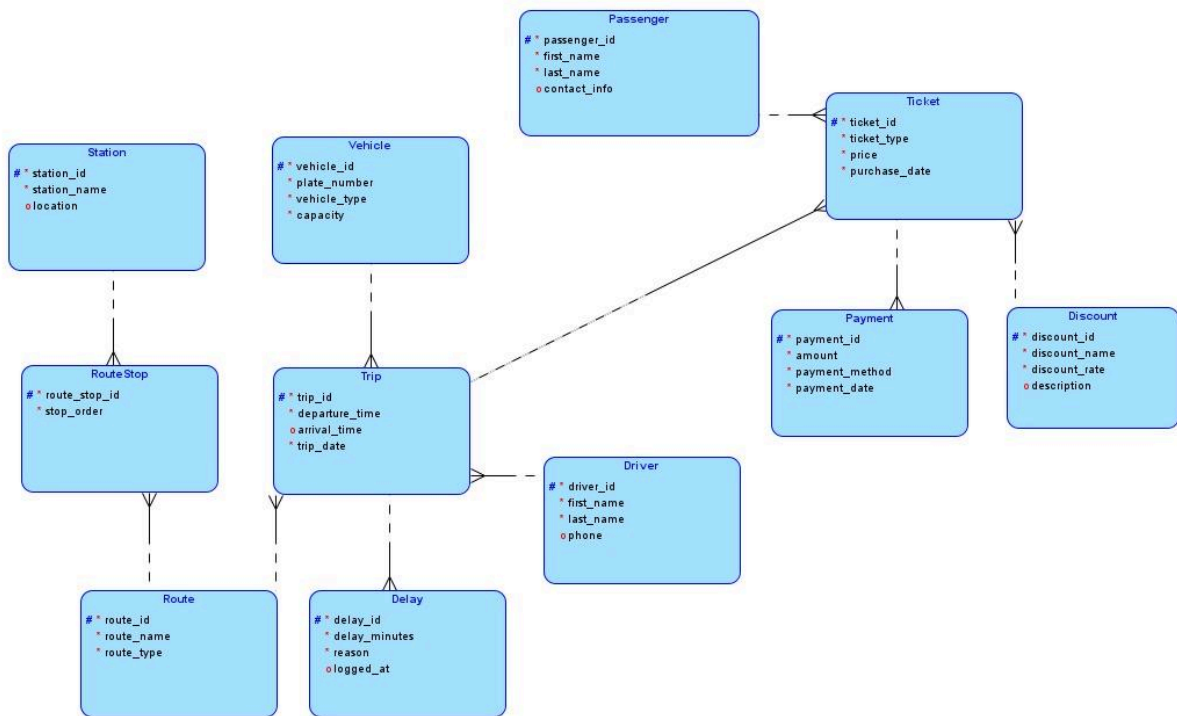
- **Data Integrity:** The system must ensure that every ticket is linked to a valid trip and a registered passenger.
- **Normalization:** The database must be designed in 3rd Normal Form (3NF) to prevent data redundancy.
- **GUI Accessibility:** A simple interface must allow staff to edit table data without complex authentication.

3. Database Schema

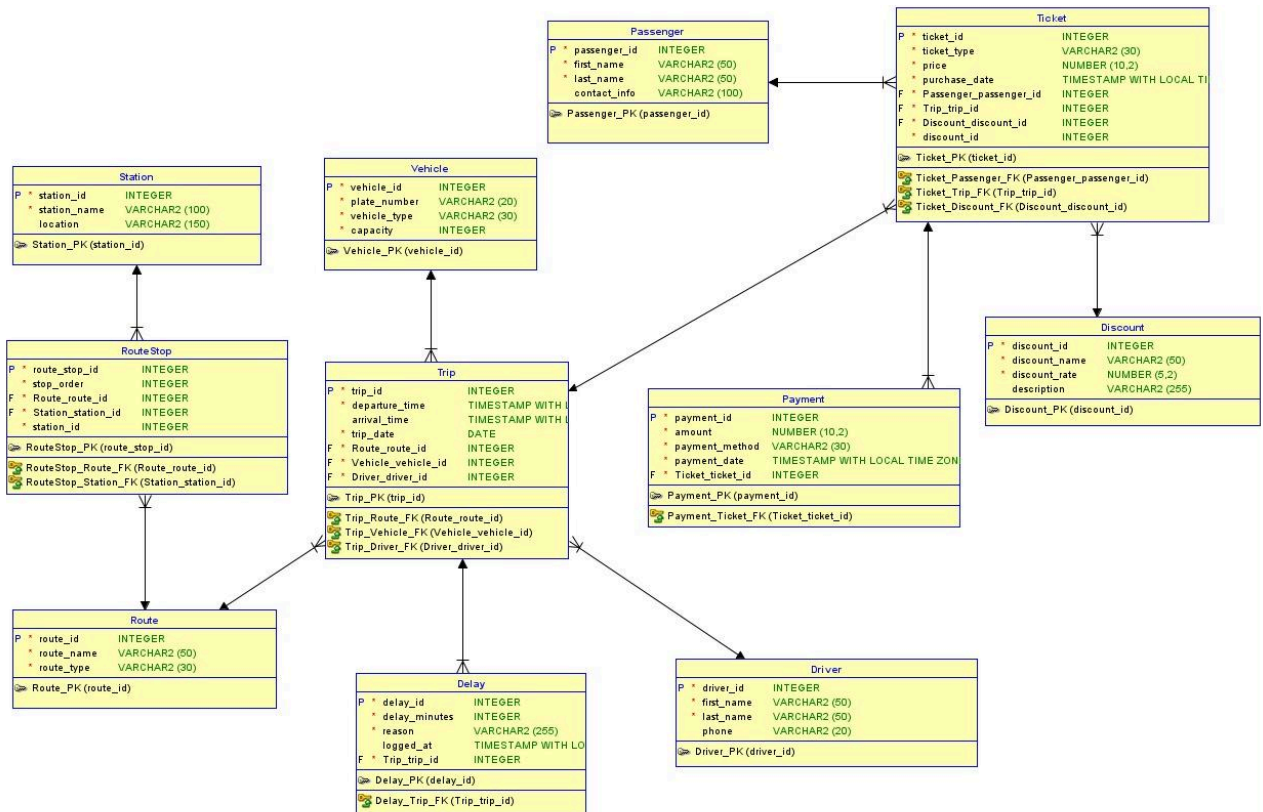
1. **Passengers:** Personal details of commuters.
2. **Stations:** Names and locations of all stops.
3. **Routes:** Definitions of transport lines (e.g., Route 10).
4. **RouteStops:** The specific sequence of stations for each route.
5. **Vehicles:** Data on buses, trams, or trains.
6. **Drivers:** Information about the teaching/operating staff.
7. **Trips:** Scheduled journeys at specific times.
8. **Tickets:** Transaction records for journey access.
9. **Payments:** Financial records of ticket sales.
10. **Discounts:** Rules for reduced fares (Student, Senior).

11. Delays: Records of time deviations from the schedule.

4. Entity Model (Logical)



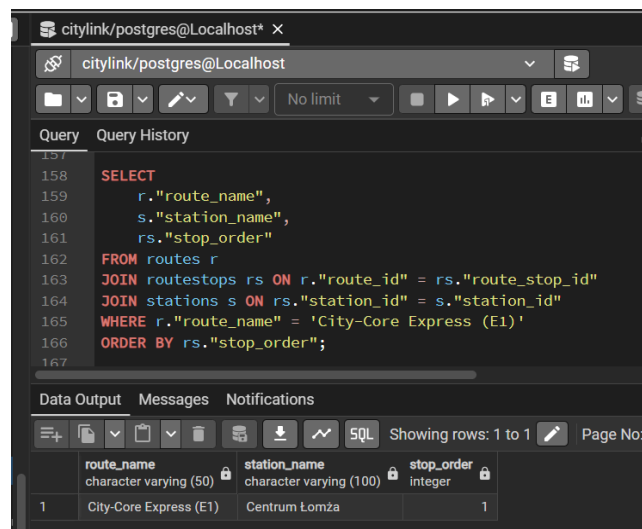
5. Relational Model



6. Implementation Description

Database System: PostgreSQL (pgAdmin4)

- **Normalization:** The database has been designed in the **3rd Normal Form (3NF)** to ensure data integrity and eliminate redundancy. A key example of this design is the **"routestops"** junction table. This table is strategically used to resolve the **Many-to-Many (M:N) relationship** between the **"routes"** and **"stations"** entities. Instead of duplicating station data within routes, the junction table stores the specific sequence of stations (**stop_order**) for each line.



- **Relational Integrity:** The implementation relies on strict Foreign Key constraints to maintain operational consistency. For instance, a **"Trip"** cannot exist without a valid **"Vehicle"** and **"Driver"**. Similarly, every **"Ticket"** is logically linked to a specific **"Passenger"** profile and a scheduled **"Trip"**.
- **Data Verification (SQL Queries):** To verify the system's logic, the following relational queries were executed in pgAdmin4:
 - **Route Mapping:** A JOIN query between routes, routestops, and stations confirms that the "City-Core Express (E1)" line correctly retrieves its designated sequence from the database.
 - **Operational Integration:** Queries linking trips, vehicles, and drivers demonstrate that real-time operational data, such as plate numbers and driver names, are correctly associated with specific journey IDs.
 - **Financial Logic:** A complex query joining tickets, passengers, and discounts verifies the automated pricing engine, which applies student or senior discounts to the base price to calculate the final fare.

Query

Query History

Scratch Pad

169

SELECT

170

tr."trip_id",

171

v."plate_number",

172

d."first_name" || ' ' || d."last_name" AS driver,

173

r."route_name",

174

tr."departure_time"

175

FROM trips tr

176

JOIN vehicles v ON tr."vehicle_id" = v."vehicle_id"

177

JOIN drivers d ON tr."driver_id" = d."driver_id"

178

JOIN routes r ON tr."route_id" = r."route_id";

Data Output

Messages

Notifications

SQL

Showing rows: 1 to 103

Page No: 1 of 1

	trip_id integer	plate_number character varying (20)	driver text	route_name character varying (50)	departure_time timestamp with time zone
1	1	EL-1001	Jan Kowal...	City-Core Express (E1)	2026-04-16 13:18:51.782588+...
2	2	EL-1002	Jan Kowal...	Harbor Bridge (H14)	2026-04-16 13:18:51.782588+...
3	3	EL-1003	Jan Kowal...	Sunset Valley (SV8)	2026-04-16 13:18:51.782588+...
4	4	EL-1001	Jan Kowal...	City-Core Express (E1)	2026-04-13 19:04:54.876748+...
5	5	EL-1002	Jan Kowal...	Harbor Bridge (H14)	2026-04-13 19:04:54.876748+...
6	6	EL-1003	Jan Kowal...	Sunset Valley (SV8)	2026-04-12 19:04:54.876748+...
7	7	EL-1004	Jan Kowal...	Hat1	2026-04-13 19:04:54.876748+...
8	8	EL-1005	Jan Kowal...	City-Core Express (E1)	2026-04-12 19:04:54.876748+...
9	9	EL-1006	Jan Kowal...	Harbor Bridge (H14)	2026-04-12 19:04:54.876748+...

Total rows: 103

Query complete 00:00:00.048

CRLF

Query

Query History

Scratch Pad

189

190

191

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197

198

199

200

201

```

SELECT
    t."ticket_id",
    p."first_name" || ' ' || p."last_name" AS passenger,
    t."price",
    ds."discount_name",
    (t."price" * (1 - ds."discount_rate"/100)) AS final_price
FROM tickets t
JOIN passengers p ON t."passenger_id" = p."passenger_id"
JOIN discounts ds ON t."discount_id" = ds."discount_id";

```

Data Output

Messages

Notifications

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SQL

Showing rows: 1 to 2

✎

Page No:

1

	ticket_id integer	passenger text	price numeric (10,2)	discount_name character varying (50)	final_price numeric
1	501	Pass1 Surname...	4.00	Student	2.00000000000000000000000000000000
2	502	Pass2 Surname2	12.00	None	12.00000000000000000000000000000000

7. Graphical User Interface (GUI)

Technology: JavaScript (React/Next.js)

The "CityLink" GUI is a comprehensive management platform that allows administrative staff to interact with the PostgreSQL database through a modern, user-friendly interface. The GUI allows administrative staff to view and edit data in the implemented tables.

CITYLINK

URBAN KINETICISM

Dashboard

Passengers

Stations

Routes

Vehicles

Drivers

Trips

Tickets

Payments

Discounts

Delays

Admin Unit 01

SYSTEM OVERLORD

Search operational data...

ACTIVE VEHICLES142LIVE

TOTAL PASSENGERS1.0kLIVE

DAILY TICKETS8,922-0.8%

ACTIVE DELAYS20Critical: 6

Passenger Volume

Route deployment and utilization from live database

DailyWeekly

Live Delays

Sunset Valley (SV8)
Door stuck
Vehicle: EL-1003 — 8 min delay

City-Core Express (E1)
Weather
Vehicle: EL-1001 — 21 min delay

Harbor Bridge (H14)
Engine Failure
Vehicle: EL-1002 — 18 min delay

VIEW HISTORY

CITYLINK

URBAN KINETICISM

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Passengers

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Delays

Admin Unit 01

SYSTEM OVERLORD

Search operational data...

PLATFORM / STATION DIRECTORY

Station Overview

Geographic transit nodes across the Lomza metropolitan network.

4Total Stations

3Active (Linked to Routes)

1Unlinked Stations

1Max Routes / Station

STATION ID	STATION NAME	LOCATION	ROUTES THROUGH	STATUS
STN-0001	Centrum Łomża	Łomża, City Center	1	Active
STN-0002	Akademia Łomżyńska	Łomża, ul. Akademicka	1	Active
STN-0003	Dworzec Autobusowy	Łomża, Al. Legionów	1	Active
STN-0006	Dworzec Lomza	18-400	0	Unlinked

CITYLINK

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Admin Unit 01

SYSTEM OVERLORD

Search operational data...

PLATFORM / ROUTE CONTROL

Route & Schedule Control

Transit line management and station sequence mapping via RouteStops junction.

ALL ROUTES (6)

City-Core Express (E1)
Express • 26 Trips

Harbor Bridge (H14)
Standard • 26 Trips

Sunset Valley (SV8)
Standard • 26 Trips

Hat1
Express • 25 Trips

City-Core Express (E1)
Bus • 0 Trips

Harbor Bridge Link (H1)
Bus • 0 Trips

STATION SEQUENCE — CITY-CORE EXPRESS (E1)

Bus3 Stops0 Total Trips

1Centrum Łomża
Łomża, City Center

2Akademia Łomżyńska
Łomża, ul. Akademicka

3Dworzec Autobusowy
Łomża, Al. Legionów

CITYLINK

URBAN KINETICISM

Dashboard

Passengers

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Discounts

Delays

AU

Admin Unit 01

SYSTEM OVERLORD

Search operational data...

Management Platform

AU

PLATFORM / DRIVER REGISTRY

Driver Registry

Operator profiles with trip history from the Trips table associations.

26

Registered Drivers

1

With Active Trips

103

Total Trips Logged

DRIVER	DRIVER ID	PHONE	TOTAL TRIPS	LAST TRIP	STATUS
JK Jan Kowalski	DRV-0001	12345	103 trips	16.04.2026	Active
MK Michal Kowalski	DRV-0002	+48 579 199 706	0 trips	No trips yet	Standby
KN Krzysztof Nowak	DRV-0003	+48 521 322 247	0 trips	No trips yet	Standby
TW Tomasz Wisniewski	DRV-0004	+48 511 727 615	0 trips	No trips yet	Standby

Search operational data...

Management Platform

AU

PLATFORM / TICKET REGISTRY

Ticket Sales

Revenue and ticket issuance records linked to Passengers and Trips.

TODAY'S TICKETS

8,922

TODAY'S REVENUE

PLN 44 615,11

TOTAL RECORDS

200

TICKET ID	PASSENGER	ROUTE	TYPE	PRICE	DISCOUNT	PURCHASED
TKT-00501	Pass1 Surname12	Harbor Bridge (H14)	Single-Trip	PLN 4.00	Student	16.04.2026
TKT-00502	Pass2 Surname2	Harbor Bridge (H14)	Daily	PLN 12.00	None	16.04.2026
TKT-00903	Pass2 Surname2	City-Core Express (E1)	Standard	PLN 6.25	—	16.04.2026
TKT-00904	Pass3 Surname3	Harbor Bridge (H14)	Standard	PLN 4.46	—	16.04.2026
TKT-00905	Pass4 Surname4	Sunset Valley (SV8)	Standard	PLN 6.97	—	16.04.2026